7.3. Integration of trig functions

$$\int \sin^n x\,dx = -\frac{1}{n}\sin^{n-1} x\cos x + \frac{n-1}{n}\int \sin^{n-2} x\,dx$$

$$\int \cos^n x\,dx = \frac{1}{n}\cos^{n-1} x\sin x + \frac{n-1}{n}\int \cos^{n-2} x\,dx$$

Example:

$n=4$

$$\int \sin^4 x\,dx = -\frac{1}{4}\sin^3 x\cos x + \frac{3}{4}\int \sin^2 x\,dx$$

$n=2$

$$= -\frac{1}{4}\sin^3 x\cos x + \frac{3}{4}\left[-\frac{1}{2}\sin x\cdot\cos x + \frac{1}{2}\int dx\right]$$

$$= -\frac{1}{4}\sin^3 x\cos x + \frac{3}{4}\left[-\frac{1}{2}\sin x\cdot\cos x + \frac{1}{2}x\right] + C$$

$$= -\frac{1}{4}\sin^3 x\cos x - \frac{3}{8}\sin x\cos x + \frac{3}{8}x + C$$

nota :

$$\int \sin^2 x\,dx = \tfrac{1}{2}\int (1-\cos 2x)\,dx = \tfrac{1}{2}x - \tfrac{1}{4}\sin 2x + C$$

$$\int \cos^2 x\,dx = \tfrac{1}{2}\int (1+\cos 2x)\,dx = \tfrac{1}{2}x + \tfrac{1}{4}\sin 2x + C$$

Also: $\sin 2x = 2\sin x \cdot \cos x$

$\cos 2x = \cos^2 x - \sin^2 x$

$= 2\cos^2 x - 1$

$= 1 - 2\sin^2 x$

INTEGRATING PRODUCTS OF SINES AND COSINES

$\int \sin^m x \cos^n x\,dx$	PROCEDURE	RELEVANT IDENTITIES
n odd	• Split off a factor of $\cos x$. • Apply the relevant identity. • Make the substitution $u = \sin x$.	$\cos^2 x = 1 - \sin^2 x$
m odd	• Split off a factor of $\sin x$. • Apply the relevant identity. • Make the substitution $u = \cos x$.	$\sin^2 x = 1 - \cos^2 x$
$\begin{cases} m \text{ even} \\ n \text{ even} \end{cases}$	• Use the relevant identities to reduce the powers on $\sin x$ and $\cos x$.	$\begin{cases} \sin^2 x = \frac{1}{2}(1-\cos 2x) \\ \cos^2 x = \frac{1}{2}(1+\cos 2x) \end{cases}$

Example :

$$I = \int \sin^4 x \cdot \cos^5 x \, dx$$

$$= \int \sin^4 x \cdot \cos^4 x \cdot \cos x \cdot dx$$

$$u = \sin x \quad \Rightarrow \quad du = \cos x \, dx$$

$$I = \int \sin^4 x \cdot (1 - \sin^2 x)^2 \cdot \cos x \, dx$$

$$I = \int u^4 \cdot (1 - u^2)^2 \cdot du$$

ingat : $(a-b)^2 = a^2 - 2ab + b^2$

$$I = \int u^4 \cdot (1 - 2u^2 + u^4) \, du$$

$$I = \int (u^4 - 2u^6 + u^8) \, du$$

$$= \frac{u^5}{5} - \frac{2u^7}{7} + \frac{u^9}{9} + C$$

$$= \frac{(\sin x)^5}{5} - \frac{2 \sin^7 x}{7} + \frac{\sin^9 x}{9} + C$$

$$I = \int \sin^4 x \cdot \cos^4 x \, dx$$

$$\begin{cases} \sin^2 x = \frac{1}{2}(1 - \cos 2x) \\ \cos^2 x = \frac{1}{2}(1 + \cos 2x) \end{cases}$$

$$= \int (\sin^2 x)^2 \cdot (\cos^2 x)^2 \, dx$$

$$= \int \left[\frac{1}{2}(1 - \cos 2x)\right]^2 \cdot \left[\frac{1}{2}(1 + \cos 2x)\right]^2 dx$$

$$= \frac{1}{16} \int (1 - \cos 2x)^2 \cdot (1 + \cos 2x)^2 \, dx$$

$$= \frac{1}{16} \int \left[(1 - \cos 2x)(1 + \cos 2x)\right]^2 dx$$

ndk: $(a-b)(a+b) = a^2 - b^2$

$$= \frac{1}{16} \int \left(1^2 - \cos^2 2x\right)^2 \cdot dx$$

$$= \frac{1}{16} \int \left(\sin^2 2x\right)^2 dx$$

$$= \frac{1}{16} \int \sin^4 2x \cdot dx$$

Sub: $u = 2x$ use the power reduction formula.

another approach ?

$$\begin{cases} \sin^2 x = \frac{1}{2}(1 - \cos 2x) \\ \cos^2 x = \frac{1}{2}(1 + \cos 2x) \end{cases}$$

$$\frac{1}{16} \int (\sin^2 2x)^2 \, dx$$

$$= \frac{1}{16} \int \left(\frac{1}{2}(1 - \cos 4x)\right)^2 dx$$

$$= \frac{1}{64} \int (1 - \cos 4x)^2 \cdot dx$$

$$= \frac{1}{64} \int (1 - 2\cos 4x + \cos^2 4x) \, dx$$